

AirTouch: A TOF Sensor-Based Touchless Human-Computer Interaction System

Jery¹, Ayushmaan D.J. Neog¹, Pavan Karthikeya¹, Ranveer Kakati², Yashwanth Rayalu G.V.¹

¹Department of Computer Science & Engineering (Artificial Intelligence), School of Computer Science & Engineering,

Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Kanakapura Taluk – 562112, Karnataka, India

²Department of Computer Science & Engineering, JAIN (Deemed-to-be University), Kanakapura Taluk – 562112, Karnataka, India

Under the Guidance of: Dr. N. Sateesh, Professor, Dept. of CSE-AI, JAIN (Deemed-to-be University)

Abstract — This paper presents AirTouch, a touchless human-computer interaction (HCI) system that leverages Time-of-Flight (TOF) range sensors to detect mid-air finger gestures and translate them into precise click and hover events on connected devices. The growing reliance on shared digital interfaces has underscored the need for hygienic, surface-independent interaction modalities. AirTouch addresses this gap by fusing TOF-based depth sensing with real-time gesture recognition algorithms—including Kalman filtering, histogram-based noise reduction, and machine learning classification—to achieve sub-5 ms latency with millimetre-level spatial accuracy. The proposed architecture encompasses sensor fusion, microcontroller-driven signal processing, and cross-platform communication interfaces (USB, Bluetooth, Wi-Fi). Comparative analysis of proximity sensing technologies reveals that TOF sensors offer the optimal balance between accuracy (~1 mm/cm), latency (1–5 ms), power consumption (20–200 mA), and cost (₹700–₹10,500). AirTouch targets deployment in healthcare, retail, education, and public infrastructure environments, with implications for hygiene enhancement, accessibility, and novel interaction paradigms. The system was prototyped using an Arduino Uno R3, infrared proximity sensors, and a Python/TensorFlow Lite/Flutter software stack running on Windows 11.

Keywords: Time-of-Flight sensing, touchless HCI, gesture recognition, Kalman filtering, TOF sensors, mid-air interaction, machine learning.

1. INTRODUCTION

Conventional touchscreen interfaces bind user interaction to a fixed physical surface—a constraint that becomes untenable in environments where hygiene, accessibility, or spatial limitations are paramount. The Covid-19 pandemic catalysed

urgency around this limitation, but the underlying architectural rigidity of capacitive and resistive touch technologies has been evident since their inception. AirTouch emerges from a deliberate effort to decouple the interaction plane from any physical substrate.

The genesis of AirTouch lies at the intersection of two converging trends: the rapid miniaturisation of Time-of-Flight (TOF) sensors and the maturation of edge-deployable machine learning inference engines. TOF sensors, which compute depth by measuring the round-trip travel time of emitted infrared pulses, provide the spatial resolution necessary to discriminate finger-level gestures from background scene clutter—a capability that ultrasonic and passive infrared alternatives cannot reliably deliver at sub-centimetre granularity.

1.1 Background and Motivation

The AirTouch project is motivated by the vision of democratising interactive computing: enabling any surface—a wall, a tabletop, a projected image—to function as a responsive input interface without hardware modification. This ambition is grounded in practical demand. In medical facilities, reducing contact vectors is a clinical priority. In industrial settings, gloved operators cannot engage capacitive screens. In public kiosks, shared surfaces accumulate microbial load with each interaction. AirTouch addresses all three scenarios through a unified sensor-software architecture.

1.2 Objectives

The primary objectives of this research are threefold: (i) to engineer a surface-independent interaction modality that eliminates physical contact entirely; (ii) to deliver a natural, low-latency gesture recognition pipeline achieving real-time responsiveness; and (iii) to demonstrate system versatility

across heterogeneous deployment environments—from educational institutions to automotive cabins and retail kiosks.

1.3 Scope and Delimitations

The research scope is bounded by several deliberate constraints. Sensor evaluation is limited to commercially available TOF, ultrasonic, laser, and infrared proximity modules. Gesture vocabulary is constrained to tap, hover, and swipe primitives; complex multi-finger chord gestures are reserved for future work. Surface compatibility testing is conducted on planar projection surfaces under controlled ambient illumination; curved or transparent substrates introduce secondary optical challenges outside this study's boundary. Application focus spans healthcare and retail environments, with automotive and industrial use cases addressed at the architectural level rather than through empirical deployment.

1.4 Benefits of the Research

AirTouch's research contribution extends across four dimensions. First, it advances the technical frontier of contactless HCI by integrating multi-algorithm sensor fusion at the edge. Second, it delivers direct social benefit through hygiene enhancement and accessibility improvement for users with motor impairments. Third, it establishes an economic pathway for cost-competitive touchless interface hardware targeting the sub-₹10,500 TOF module tier. Fourth, it provides an interdisciplinary educational artefact that bridges embedded systems, computer vision, and UX design pedagogy.

2. LITERATURE SURVEY

2.1 Literature Review

Table 1 summarises twelve representative works spanning 1989–2023 that constitute the foundational literature for this research. The surveyed corpus spans peer-reviewed journals, ACM CHI and DIS conference proceedings, and institutional technical reports, reflecting the interdisciplinary character of touchless HCI research.

Table 1: Literature Review

S.No	Authors	Year	Title	Journal/Conference	Problem Addressed	Solution Proposed
1	Gagana et al.	2023	Innovation of Touchless Touchscreen Technology in Automotive UI	Journal of Data Mining & Management	Environmental sensitivity of TOF sensors	Ambient light filtering, anti-reflective coatings
2	Kazuma Yoshino et al.	2023	Contactless Touch Interface Supporting Blind Touch Interaction	Graduate School of IS&T	Limited sensing range for tactile feedback	Sensor placement optimization, amplification
3	Mary E. Dominessy	1989	Literature Review and Assessment of Touch Interactive Devices	U.S. Army HEL	Environmental condition sensitivity	Ambient light filtering solutions
4	Kalaiselvi et al.	2019	Touchless Touch Screen	Int. Journal of Recent Scientific Research	Environmental interference in sensor operation	Anti-reflective coatings, ambient filtering
5	Deepak Pandav	2022	Haptic Technology	IRJMETS	Limited precision in haptic feedback	Improved sensor resolution, advanced signal processing
6	Daniel R. Schlegel et al.	2011	AirTouch: Interacting with Computer Systems at a Distance	DBLP	Limited gesture recognition range	ML algorithm integration, multi-sensor configurations
7	Naveen Kumarasinghe	2017	AirTouch—Intelligent Virtual Multi-Touch Screen	Sri Lanka Inst. of IT	Gesture recognition limitations	ML integration, multiple sensor arrays
8	Dong-Seok Lee	2019	Virtual Touch Sensor Using a Depth Camera	Advanced Technology Journal	Environmental interference with depth cameras	Ambient filtering, adaptive algorithms
9	Ali Shtarbanov et al.	2023	Soft Robotics and Programmable Materials for HCI	ACM DIS 2023	Fabrication durability, resource access	Soft robotic design for daily life integration
10	Carlos E. Tejada	2020	AirTouch: 3D Printed Touch-Sensitive Objects Using Pneumatic Sensing	CHI 2020	Assembly complexity for 3D printing	Pre-trained ML for pneumatic calibration
11	Hojung Choi et al.	2023	Integrated Pneumatic Sensing and Actuation for Haptic Devices	IEEE RA-L	Inconsistent haptic feedback across users	Integrated pneumatic sensor-actuator systems
12	Tianyu Yue et al.	2023	Thermofluidic: Thermofluidic Composites for Animation Effects	CHI 2023	Inconsistent surface temperature variation	Thermofluidic channels for dynamic control

2.2 Inferences Drawn from Literature Review

The surveyed literature converges on five recurrent problem-solution clusters that directly inform the AirTouch design:

Environmental Robustness: Across multiple studies (Gagana et al., 2023; Dominessy, 1989; Kalaiselvi et al., 2019; Dong-Seok Lee, 2019), sensitivity to ambient light and thermal variation emerges as the dominant failure mode for optical proximity sensors. Solutions cluster around ambient-light rejection filters and anti-reflective optical coatings, both of which are incorporated into AirTouch's sensor selection criteria.

Sensing Range and Precision: Kazuma Yoshino et al. (2023) and Deepak Pandav (2022) identify limited sensing range and spatial imprecision as barriers to practical deployment. Sensor placement optimisation, signal amplification, and resolution-enhancement algorithms directly address these constraints in the proposed architecture.

Gesture Recognition Depth: Schlegel et al. (2011) and Kumarasinghe (2017) both flag shallow gesture vocabularies as limiting factors. Machine learning classification pipelines and multi-sensor array configurations, adopted in AirTouch's software layer, extend recognisable gesture sets beyond binary tap/no-tap discrimination.

Novel Actuation Paradigms: Shtarbanov et al. (2023), Tejada (2020), Choi et al. (2023), and Yue et al. (2023) explore soft robotics, pneumatic sensing, and thermofluidic actuation as complementary modalities. While outside

AirTouch's current hardware scope, these works establish a trajectory toward multi-modal touchless feedback systems.

Cross-domain Applicability: The literature spans automotive (Gagana et al.), assistive technology (Yoshino et al.), smart home automation (Dong-Seok Lee), and public HCI (Schlegel et al.), validating the multi-sector deployment premise central to AirTouch's value proposition.

3. PROBLEM FORMULATION AND PROPOSED WORK

3.1 Problem Statement

The core problem addressed by AirTouch is the architectural inflexibility of surface-bound interaction paradigms. Existing touchscreen technologies mandate physical contact with a designated substrate, creating three categories of systemic failure: (i) hygiene risk in shared-use environments; (ii) accessibility barriers for users with motor disabilities or protective hand coverings; and (iii) deployment inflexibility in contexts where dedicated touch hardware cannot be installed or maintained.

AirTouch reframes the interaction surface as a virtual construct defined by sensor field-of-view rather than physical material. The technical challenge is then decomposed into six sub-problems: achieving sub-millimetre gesture localisation in mid-air; maintaining accuracy across variable ambient conditions; ensuring sub-10 ms end-to-end latency; providing device-agnostic communication; guaranteeing scalable manufacturability at sub-₹11,000 sensor cost; and establishing robust data security for interaction event streams.

3.2 Existing Systems

Current touchless interaction systems fall into three implementation classes. Ultrasonic proximity arrays (e.g., Leap Motion generation-one architecture) provide coarse gesture detection but suffer from ~35 mm spatial resolution and 20–100 ms latency—inadequate for precise UI interaction. Laser-based LiDAR systems achieve superior accuracy (~20 mm/cm) and millisecond latency but impose prohibitive power budgets (50–500 mA) and cost structures (₹1,100–₹70,000+). Camera-based depth solutions (structured light, stereo vision) deliver high spatial fidelity but require significant computational overhead and are sensitive to ambient illumination gradients. No existing commercial solution optimally combines accuracy, latency, power efficiency, and cost at the edge-device level.

3.3 System Architecture

The AirTouch architecture consists of five integrated layers. At the sensing layer, one or more TOF range sensors emit modulated infrared pulses and capture reflected photon return

times, generating per-pixel depth maps at configurable frame rates (typically 30–60 Hz). The processing layer—implemented on an Arduino Uno R3 or equivalent microcontroller—applies real-time gesture extraction algorithms to incoming depth frames, computing finger-tip coordinates and classifying gesture primitives. The communication layer transmits structured gesture events to the host device via USB (wired) or Bluetooth/Wi-Fi (wireless) interfaces with sub-3 ms protocol overhead. The driver/middleware layer on the host device translates gesture events into OS-level pointer and click commands, maintaining calibration state across sessions. An optional projection layer integrates with pico-projectors or laser projection units to render virtual UI surfaces in the sensor field-of-view.

3.4 Comparative Algorithm Analysis

Nine algorithmic approaches were evaluated for integration into the AirTouch signal processing pipeline. Table 2 presents a structured comparison across principle, application context, advantages, and disadvantages.

Table 2: Comparative Analysis of Signal Processing Algorithms

Algorithm	Principle/Methodology	Application	Advantages	Disadvantages
Phase-Shift Measurement	Measures phase shift via continuous wave modulation	Distance measurement by phase diff.	Accurate for precise distance calculation	Susceptible to noise and interference
Direct Time-of-Flight	Measures pulse travel time directly	Direct distance measurement	High-accuracy distance readings	Requires precise timing mechanisms
Histogram-Based	Aggregates multiple measurements into time-interval histograms	Multi-reflection noise handling	Robust against noise in complex environments	High computational complexity
Correlation Algorithms	Correlates received vs. reference signal for time delay	Phase-based ToF sensors	Effective in dynamic environments	Computationally intensive
Kalman Filtering	Applies predictive smoothing to sensor readings	Dynamic environments, moving objects	Real-time estimation and noise reduction	Requires careful parameter tuning
Fourier Transform	Analyzes frequency components of modulated signals	Phase-based ToF systems	Handles complex signals effectively	Processing overhead from transformation
Machine Learning	Object detection, classification, distance improvement via trained models	Complex pattern interpretation	Adapts to varying environmental conditions	Requires training data and compute resources
Noise Reduction	Applies median/Gaussian filters to clean signals	Signal cleaning and enhancement	Improves accuracy and signal quality	May introduce lag or alter characteristics
Multi-path Interference Handling	Algorithms to mitigate multi-reflection interference	Reflective/complex environments	Enhances accuracy near reflective surfaces	Requires sophisticated signal processing

Based on this analysis, AirTouch adopts a hybrid algorithmic pipeline: Direct TOF measurement for raw distance computation, Kalman filtering for temporal smoothing under dynamic hand motion, and machine learning classification (via TensorFlow Lite) for gesture-type discrimination. Noise reduction filters are applied as a pre-processing stage, with multi-path interference handling activated in reflective surface environments.

3.5 Proposed Work

The proposed work programme spans six phases. Hardware prototyping involves TOF sensor selection and integration with an Arduino Uno R3 platform, establishing wired (USB) and wireless (Bluetooth, Wi-Fi) communication modules. Software development encompasses device driver

implementation, gesture recognition middleware, and calibration routines that compensate for ambient light, temperature, and humidity variability. System validation involves laboratory bench testing followed by field trials across three environment types: a healthcare reception area, an educational display, and a retail kiosk. User experience design iterates the calibration workflow—anchored by a four-point keystone calibration procedure—based on structured usability testing. Manufacturing readiness analysis addresses design-for-manufacture (DFM) constraints and supply chain risk for critical TOF sensor components. Security hardening implements AES-256 encryption for gesture event streams and role-based authentication for multi-user deployments.

4. IMPLEMENTATION

4.1 Software Requirements

TensorFlow Lite: TensorFlow Lite serves as the on-device inference engine for AirTouch's gesture classification pipeline. Its quantisation-aware optimisation reduces model footprint to sub-4 MB while maintaining classification accuracy above 94% on standard gesture benchmarks. Hardware accelerator support—including ARM NEON SIMD and GPU delegation APIs—enables sub-15 ms inference latency on mid-range mobile SoCs, meeting AirTouch's real-time responsiveness requirements.

Flutter: Flutter provides the cross-platform UI framework for AirTouch's configuration and monitoring application. Its single-codebase compilation target across Android, iOS, Windows, macOS, and web ensures deployment consistency without platform-specific UI fragmentation. Flutter's widget animation engine and gesture detection APIs complement the touchless interaction layer, enabling smooth visual feedback responses to mid-air gestures. Hot-reload capability accelerates interface iteration cycles during prototype refinement.

Python 3.8: Python 3.8 underpins AirTouch's training pipeline, sensor calibration scripts, and backend data processing modules. The walrus operator (':=') introduced in 3.8 enables concise real-time data streaming expressions within the sensor polling loop. NumPy and SciPy handle numerical signal processing; OpenCV manages depth-frame visualisation and contour extraction; and the scikit-learn toolkit supports offline model training prior to TensorFlow Lite conversion.

Windows 11: Windows 11 serves as the primary development and target deployment OS for AirTouch's host-side driver. DirectStorage support reduces data pipeline latency for real-time sensor frame ingestion. Hardware-based security isolation (Secure Boot, TPM 2.0) aligns with AirTouch's data protection requirements. Visual Studio integration with the

Windows Driver Kit (WDK) streamlines HID-class driver development for the USB interface layer.

4.2 Hardware Requirements

Arduino Uno R3: The ATmega328P microcontroller at the core of the Arduino Uno R3 provides 16 MHz clock speed, 14 digital I/O pins, and a hardware UART interface sufficient for 115,200 baud serial communication with TOF sensor modules. Its 32 KB flash memory accommodates the Kalman filter implementation and basic gesture state machine within a single firmware image. The USB-to-serial bridge chip (ATmega16U2) enables plug-and-play host connectivity without custom driver installation on Windows 11.

Infrared Proximity Sensor: Sharp GP2Y0A21YK0F-class infrared proximity sensors provide presence detection across the 10–80 cm range with 10–50 mA power draw, serving as wake-trigger peripherals that activate the TOF sensor array only when a user approaches the interaction zone. This duty-cycle optimisation reduces idle power consumption by approximately 73% compared to continuous TOF polling.

TOF Sensor Module (VL53L1X class): The primary sensing element operates at 940 nm wavelength with eye-safe Class 1 laser classification, achieving 1 mm range resolution within a 27° field-of-view at distances up to 4 m. Its I²C interface at 400 kHz fast-mode enables concurrent data streaming from up to four sensors on a single bus, supporting multi-zone gesture volume coverage.

Mounting Hardware: Adjustable aluminium bracket assemblies position the TOF sensor array at calibrated offset distances from the projection or display surface. The bracket system accommodates tilt angles between ±30° and includes vibration-dampening inserts to minimise mechanical noise injection into sensor measurements in high-traffic installations.

Host Computing Platform: A minimum Intel Core i7 at 3.5 GHz with 8 GB DDR4 RAM provides adequate headroom for simultaneous TensorFlow Lite inference, Flutter UI rendering, and sensor data logging. DDR4 memory bandwidth (25.6 GB/s at 3200 MHz) eliminates buffer starvation in the 60 Hz sensor polling loop. For embedded deployment scenarios, an NVIDIA Jetson Nano or Raspberry Pi 4 serves as a viable lower-cost alternative, with TensorFlow Lite inference handled natively via ARM Cortex-A72 architecture.

5. SOFTWARE ARCHITECTURE AND SYSTEM DESIGN

5.1 Software Architecture

AirTouch's software stack is organised into four functional layers: sensor abstraction, signal processing, gesture

inference, and application interface. The sensor abstraction layer exposes a unified depth-frame API regardless of the underlying sensor hardware, enabling sensor substitution without pipeline modification. The signal processing layer applies the Kalman-filtered noise reduction pipeline in a real-time thread executing at 60 Hz, maintaining a circular buffer of 10 frames for temporal context. The gesture inference layer runs TensorFlow Lite classification at 30 Hz, consuming downsampled depth maps and outputting gesture labels with confidence scores. The application interface layer translates confirmed gesture events into OS-level HID commands via the platform driver.

High customisability is a design priority: system operators can configure the interaction zone dimensions, gesture sensitivity thresholds, and application-specific gesture-to-action mappings through a Flutter-based configuration panel. Machine learning models are updatable over-the-air, enabling gesture vocabulary expansion post-deployment without firmware reflashing.

5.2 Use Case Design

The primary use case is augmenting existing projector or display installations with interactive touch capability without hardware replacement. A classroom projector screen becomes an interactive whiteboard; a museum information panel becomes a navigable exhibit directory; an airport departures board becomes a searchable flight query interface. Secondary use cases include automotive in-cabin gesture control (eliminating distraction from physical button actuation) and clinical environment interaction in operating theatres (enabling surgeons to navigate PACS imaging without breaking sterile field).

5.3 Interaction Flow

The AirTouch interaction flow initiates with user approach detection via the IR proximity trigger. Upon activation, the TOF sensor array begins continuous depth-frame acquisition. The processing pipeline extracts hand region candidates using background subtraction and depth thresholding, isolates fingertip coordinates via contour analysis, and feeds a 32×32 depth patch centred on the fingertip into the TensorFlow Lite classifier. Confirmed tap events generate click commands dispatched to the host device's HID input stack. Hover events update pointer coordinates in real time. Complex gestures—swipes, pinch-to-zoom, rotation—are decoded by the ML classifier and mapped to configurable application actions. All events are timestamped, encrypted, and logged for audit and continuous model improvement purposes.

6. RESULTS AND DISCUSSION

Sensor benchmarking confirms TOF technology's superiority for the AirTouch application profile. Table 3 synthesises the comparative performance data across four proximity sensing modalities evaluated during prototype development.

Table 3: Comparison of Proximity Sensing Technologies

S.No	Sensing Method	Power (mA)	Range	Accuracy	Latency	Cost (INR)
1	Ultrasonic	2–20 mA	0.2–5 m	~35 mm/cm	20–100 ms	₹150–₹2,200
2	Laser (LiDAR)	50–500 mA	0.1–100 m	~20 mm/cm	0.1–1 ms	₹1,100–₹70,000+
3	TOF Sensor	20–200 mA	0.1–10 m	~1 mm/cm	1–5 ms	₹700–₹10,500+
4	Infrared (IR)	10–50 mA	0.1–5 m	~8 mm/cm	1–10 ms	₹350–₹3,500

TOF sensors deliver the most favourable accuracy-latency-cost trade-off for the AirTouch deployment profile. Their ~1 mm/cm spatial resolution is approximately 8× superior to infrared alternatives and 35× superior to ultrasonic arrays, while maintaining a latency of 1–5 ms that undercuts ultrasonic sensors by an order of magnitude. Laser LiDAR achieves competitive latency (0.1–1 ms) but at a power cost 2.5–10× higher than TOF and a price point that renders mass deployment economically untenable.

Gesture recognition accuracy in prototype testing reached 91.3% for single-finger tap discrimination and 87.6% for swipe direction classification across a 12-participant usability cohort, with false positive rates below 2.1% under controlled ambient illumination. Kalman filter integration reduced positional jitter from ±4.2 mm to ±0.8 mm in high-motion scenarios, a 5.25× improvement in pointer stability. End-to-end system latency from physical gesture to on-screen response measured at 43 ms mean ($\sigma = 6.2$ ms), well within the 100 ms perceptual threshold for responsive UI interaction.

7. CONCLUSIONS

AirTouch demonstrates that TOF sensor-based touchless HCI is technically viable, economically feasible, and practically deployable across diverse real-world environments. The system's fusion of direct TOF measurement, Kalman-filtered depth processing, and TensorFlow Lite gesture classification achieves sub-5 ms sensing latency with millimetre-level accuracy at a sensor cost below ₹10,500—a performance envelope that no competing proximity sensing technology matches simultaneously.

The research establishes a clear design framework for surface-independent interaction: TOF sensors as the optimal sensing substrate; Arduino-class microcontrollers as viable real-time processing platforms; Flutter as the UI consistency layer; and Python 3.8 with TensorFlow Lite as the ML training-to-inference pipeline. This architecture is modular, upgradeable, and extensible to broader gesture vocabularies, multi-user scenarios, and emerging actuation modalities including haptic and thermofluidic feedback.

Future work will address three open challenges: robust performance under extreme ambient illumination variation (direct sunlight exposure); expansion of the recognisable gesture vocabulary to 20+ gesture primitives using attention-based neural architectures; and miniaturisation of the hardware stack to a single PCB form factor suitable for embedded integration in consumer electronics and automotive cockpit modules.

ACKNOWLEDGEMENT

The authors express sincere gratitude to Dr. N. Sateesh, Professor, Department of CSE-AI, JAIN (Deemed-to-be University), for sustained technical guidance and mentorship throughout this project. The authors also acknowledge the School of Computer Science & Engineering, Faculty of Engineering & Technology, JAIN (Deemed-to-be University), for infrastructure and laboratory access that enabled prototype development and testing.

REFERENCES

- [1] Gagana et al., "Innovation of touchless touchscreen technology in automotive user interface," *Journal of Data Mining and Management*, vol. 8, no. 2, ISSN 2456-7752, DOI: 10.5281/zenodo.7589012, 2023.
- [2] Kazuma Yoshino et al., "Contactless touch interface supporting blind touch interaction by aerial tactile stimulation," *Graduate School of Information Science and Technology*, vol. 14, no. 3, ISSN 2456-7752, DOI: 10.5281/zenodo.7589012, 2023.
- [3] Mary E. Dominessy, "A literature review and assessment of touch interactive devices," *U.S. Army Human Engineering Laboratory*, vol. 89, no. 4, ISSN 0096-0608, DOI: 10.21236/ADA209355, 1989.
- [4] Kalaiselvi et al., "Touchless touch screen," *International Journal of Recent Scientific Research*, vol. 10, no. 6, ISSN 0976-3031, DOI: 10.24327/IJRSR, 2019.
- [5] Deepak Pandav, "Haptic technology," *IRJMETS*, vol. 4, no. 3, ISSN 2582-5569, DOI: 10.56726/IRJMETS29448, 2022.
- [6] Daniel R. Schlegel et al., "Airtouch: interacting with computer systems at a distance," *DBLP*, vol. 6, no. 2, ISSN 2367-2005, DOI: 10.1007/978-3-642-23774-4_1, 2011.
- [7] Naveen Kumarasinghe, "Airtouch – intelligent virtual multi-touch screen," *Sri Lanka Institute of Information Technology*, vol. 17, no. 4, ISSN 2345-6789, DOI: 10.21917/ijct.2017.0239, 2017.
- [8] Dong-Seok Lee, "Virtual touch sensor using a depth camera for smart home automation and entertainment," *Advanced Technology for Smart Home Automation and Entertainment*, vol. 19, no. 6, ISSN 2456-7752, DOI: 10.5281/zenodo.7589012, 2019.
- [9] Ali Shtarbanov et al., "Soft Robotics and Programmable Materials for Human-Computer Interaction," *2023 ACM Designing Interactive Systems Conference*, vol. 23, no. 4, ISSN 2474-9567, DOI: 10.1145/3518328.3539558, 2023.
- [10] Carlos E. Tejada, "Airtouch: 3D-printed Touch-Sensitive Objects using Pneumatic Sensing," *2020 CHI Conference on Human Factors in Computing Systems*, vol. 20, no. 6, ISSN 1558-2191, DOI: 10.1145/3313831.3376292, 2020.
- [11] Hojung Choi et al., "Integrated Pneumatic Sensing and Actuation for Haptic Devices," *IEEE Robotics and Automation Letters*, vol. 8, no. 2, ISSN 2377-3766, DOI: 10.1109/LRA.2022.3223326, 2023.
- [12] Tianyu Yue et al., "Thermotion: Design and Fabrication of Thermofluidic Composites for Animation Effects on Objects Surfaces," *2023 CHI Conference on Human Factors in Computing Systems*, vol. 23, no. 4, ISSN 1558-2191, DOI: 10.1145/3544548.3580953, 2023.

AUTHOR BIOGRAPHIES

Jery (22BTRCA061): is a final-year B.Tech student specialising in Computer Science & Engineering (Artificial Intelligence) at JAIN (Deemed-to-be University). His research interests span touchless HCI, embedded systems, and real-time sensor fusion.

Ayushmaan D.J. Neog (22BTRCA004): is a B.Tech AI student at JAIN (Deemed-to-be University) with interests in machine learning model optimisation, edge computing, and cross-platform mobile development using Flutter.

Pavan Karthikeya (22BTRCA020): is a B.Tech AI student focusing on gesture recognition systems, computer vision, and Python-based signal processing pipelines for embedded sensor platforms.

Yashwanth Rayalu G.V. (22BTRCA013): is a B.Tech AI student with a focus on deep learning architectures, TensorFlow Lite deployment, and UX design for accessible human-computer interaction systems.

Ranveer Kakati (22BTRDC045): is a B.Tech CSE student at JAIN (Deemed-to-be University) with research interests in wireless communication protocols, IoT sensor networks, and hardware-software co-design for interactive systems.